

What if one microservice changes its API overnight... and silently breaks your entire system?



The Problem



- Independent microservices can break each other after API changes.
- Traditional integration testing often detects issues too late.



What is Contract Testing?



- Verifies the agreement (contract) between service provider and consumer.
- Ensures both services follow the same request and response format.



Consumer vs Provider



- Consumer: The service requesting data (e.g., Order Service).
- Provider: The service providing data (e.g., User/Auth Service).



What is a Contract?



- A JSON file describing expected requests and responses.
- Acts as a shared agreement between services.



How Pact Works



- Consumer generates a contract.
- Provider validates itself against that contract.
- If validation passes, both services are compatible.



Why Not Just Integration Tests?



- Faster feedback during development.
- No need to run the entire microservice ecosystem.
- Detects breaking API changes early.



Benefits



- Prevents unexpected production failures.
- Enables independent deployments.
- Improves developer confidence.
- Supports CI/CD automation.



Real-World Use Cases



- E-commerce (Order ↔ Product Service)
- Banking (Payment ↔ Account Service)
- Healthcare (Patient ↔ Appointment Service)
- Ride Sharing (Booking ↔ Driver Service)



Best Practices

